ALGEBRA & GRAPHS ·

Functions
Part 1 — Notation, Domain & Range

Function notation, mapping diagrams, domain & range, evaluating and sketching functions.

Use these notes well and you will be able to:

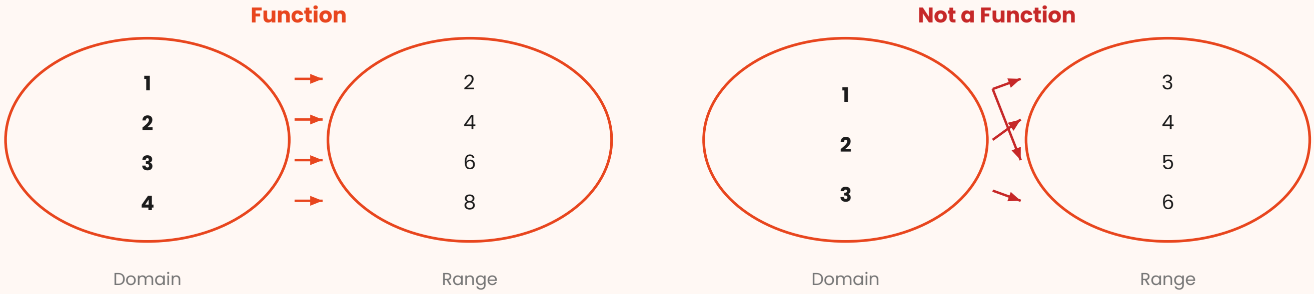
- Understand function notation f(x) andevaluate functions for given inputs.
- Identify the domain (inputs allowed) and range (outputs produced).
- Distinguish between a function and a non-function using the vertical line test.
- Draw and interpret mapping diagrams.
- Sketch and describe linear and quadratic functions including key features.

If you follow the steps, you will succeed.

What Is a Function?

A function is a rule that maps EVERY input to EXACTLY ONE output. Each x-value gives one and only one y-value.

Mapping Diagrams — Function vs Non-Function



Key Words

KEY WORD	SIMPLE DEFINITION
Function f(x)	A rule mapping each input x to exactly ONE output. Read as 'f of x'.
Input / argument	The x-value fed INTO the function. Also called the argument.
Output / image	The value the function produces. f(3) means 'output when x=3'.
Domain	The SET of all ALLOWED input values. e.g. x , x ≥ 0.
Range	The SET of all POSSIBLE output values produced by the domain.
Mapping	A rule connecting each domain element to a range element.
One-to-one	Each input gives a different output. e.g. f(x) = 2x.
Many-to-one	Multiple inputs give the same output. e.g. $f(x) = x^2$ (1 and -1 both give 1).
Vertical line test	A graph represents a function if NO vertical line crosses it more than once.
Notation f:xy	Alternative function notation. f maps x to y. e.g. f: x 3x-1.

Evaluating Functions

To evaluate f(a): substitute x = a into the function rule and simplify.

$f(x) = 3x^2 - 2x + 1$. Find $f(-2)$ and $f(0)$.

$f(-2) = 3(-2)^2 - 2(-2) + 1 = 3(4) + 4 + 1 = 12 + 4 + 1$
$f(0) = 3(0)^2 - 2(0) + 1 = 0 - 0 + 1$
$f(-2) = 17$ $f(0) = 1$

$g(x) = (2x + 3) / (x - 1)$. State the value of x for which g is undefined.

g is undefined when denominator = 0: $x - 1 = 0$
$x = 1$ is excluded from the domain. Domain: x , $x \neq 1$

Domain and Range — How to Find Them

FUNCTION TYPE	DOMAIN RESTRICTION	RANGE RESTRICTION
f(x) = linear	All real numbers (x)	All real numbers
$f(x) = x^2$	All real numbers	$f(x) \geq 0$ (never negative)
$f(x) = \sqrt{x}$	$x \geq 0$ (no neg. under √)	$f(x) \geq 0$
$f(x) = 1/x$	$x \neq 0$ (denom ≠ 0)	$f(x) \neq 0$ (never reaches 0)
$f(x) = 1/(x-a)$	$x \neq a$	$f(x) \neq 0$
$f(x) = \sqrt{(x-a)}$	$x \geq a$	$f(x) \geq 0$

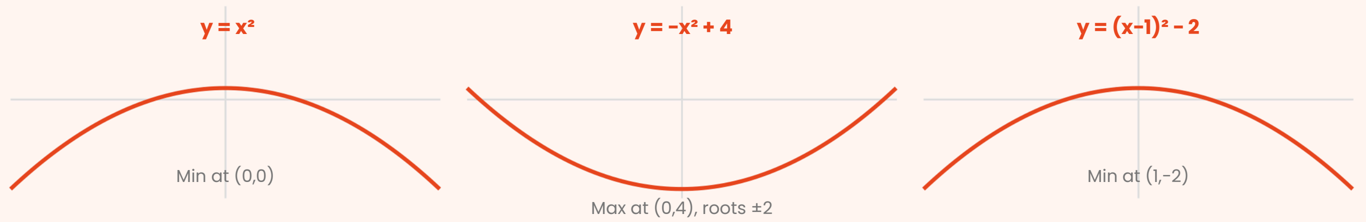
State the domain and range of $f(x) = \sqrt{(x - 3)} + 2$.

Domain: $x - 3 \geq 0$ $x \geq 3$
Range: $\sqrt{(x-3)} \geq 0$, so $f(x) = \sqrt{(x-3)}+2 \geq 2$
Domain: $x \geq 3$ Range: $f(x) \geq 2$

Sketching Key Functions — Feature Checklist

FUNCTION	y-INTERCEPT	x-INTERCEPT(S)	TURNING PT	SYMMETRY
$y = mx+c$	(0, c)	(-c/m, 0)	None	None
$y = ax^2+bx+c$	(0, c)	0, 1, or 2	$x=-b/2a$	$x = -b/2a$
$y = a/x$	None	None	None	Lines y=x, y=-x
$y = a$	(0, 1)	None	None	None. Asymp y=0
$y = x^3$	(0, 0)	(0, 0)	Point of infl.	Origin symmetry

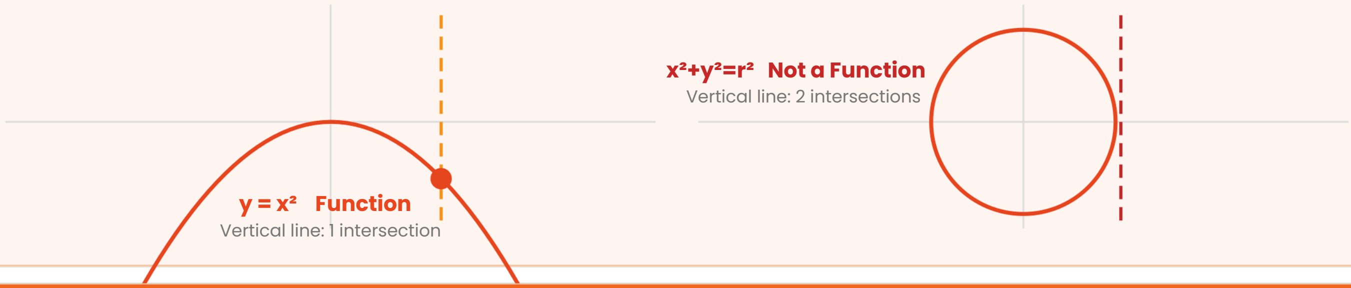
Parabola Shapes — $y = ax^2 + bx + c$



The Vertical Line Test

A graph represents a FUNCTION if and only if every vertical line crosses it AT MOST ONCE. If any vertical line crosses the graph twice or more, it is NOT a function.

Vertical Line Test



State the domain and range of f(x) = 4 - x².

This is a downward parabola, maximum at (0, 4).
Domain: all real numbers (x)
Maximum output = 4, extends down to -∞
Domain: x   Range: f(x) ≤ 4

Sketch y = 2x - 3 and state domain and range.

Gradient = 2, y-intercept = -3. Straight line, no restrictions.
x-intercept: 0 = 2x-3 x = 1.5. Points: (0,-3) and (1.5, 0).
Domain: x   Range: f(x)

Practice Questions

NOTATION AND EVALUATING

1. $f(x) = 5x - 3$. Find f(2), f(-1),and f(0).

2. $g(x) = x^2 + 4x - 7$. Find g(-3) and g(a).

3. $h(x) = 1/(2x+4)$. State the value excluded from the domain.

DOMAIN AND RANGE

4. State the domain and range of $f(x) = \sqrt{(2x - 6)}$.

5. State the domain and range of $f(x) = 3/x$.

6. A function is defined by f: x $x^2 - 9$ for $-4 \le x \le 4$. Find the range.

SKETCHING AND VLT

7. Sketch $y = -2x + 5$ labelling intercepts and gradient.

8. Sketch $y = x^2 - 6x + 5$. Find the vertex and roots.

9. Use the vertical line test to decide: is the graph of $y^2 = x$ a function?

ADD YOUR NOTES HERE

Hey smart student!

Afunction givesONE output for every input — no exceptions, ever.
Always ask: domain restriction? (divide by zero, square root of negative?)

Range follows from the shape: parabola up range ≥ minimum.

Keep going — see you in the next lesson! From FREDi